

TELONE CENTRE FOR LEARNING

TECHNICAL DEPARTMENT

DIPLOMA IN SOFTWARE ENGINEERING



AI DRIVEN ZERO TRUST CYBER SECURITY FRAMEWORK.

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CHAPTER 1: INTRODUCTION

1.1 Background of the Study

Organizations increasingly rely on interconnected digital systems, cloud computing, and remote access technologies. This transformation has expanded the attack surface for cyber threats including insider attacks, ransomware, and data breaches.

Traditional perimeter-based security models assume trust within internal networks, which is no longer valid. Zero-Trust Architecture eliminates this assumption by enforcing strict identity verification at every access point.

Artificial Intelligence enhances cybersecurity by enabling behavioral analysis and real-time anomaly detection, making it a critical component in modern security frameworks.

1.2 Problem Statement

Many organizations continue to rely on outdated security models that cannot effectively detect insider threats or respond to evolving cyberattacks in real-time.

These limitations result in increased vulnerability to data breaches, financial losses, and operational disruptions.

1.3 Aim of the Project

The aim of this project is to design and implement an AI-driven Zero-Trust cybersecurity framework that enhances real-time threat detection and automated response mechanisms.

1.4 Objectives of the Project

- To design a Zero-Trust security architecture.

- To implement AI-based anomaly detection.
- To simulate cyberattack scenarios.
- To develop automated response mechanisms.

1.5 Research Questions

- How can AI improve Zero-Trust security models?
- What types of threats can be detected using anomaly detection?
- How effective is automated response in mitigating cyber threats?

1.6 Scope of the Project

The project focuses on organizational IT environments using simulated datasets to model user behavior and cyber threats. It does not include full-scale real-world deployment.

1.7 Significance of the Study

This study contributes to cybersecurity by demonstrating how AI and Zero-Trust can be integrated to improve detection accuracy and response speed.

It provides a scalable solution applicable to various organizations.

1.8 Project Structure Overview

Chapter 1 introduces the study.

Chapter 2 reviews existing literature.

Chapter 3 presents system design.

Chapter 4 covers implementation.

Chapter 5 discusses testing and results.

Chapter 6 concludes the study.

Chapter 7 provides recommendations and technical depth appropriate for distinct ion-level work.